

Geofinitism and the Language of Light:
Measurement, Distinction and Dynamical Systems

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Pensée

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Preface

Science advances by refining the language with which measurements are recorded and shared. Every experimental result arrives first as a finite distinction registered at the limit of an apparatus. The symbols we then assign to those distinctions, photon, wavelength, emission line, interference fringe, become the working vocabulary of an entire field. When that vocabulary is treated as a set of static nouns rather than as compressions of dynamical processes, paradoxes appear that are not features of the measurements themselves but artefacts of the language used to describe them.

This *Pensée* examines the contemporary language of light as a case study. It presents the standard framing of photons, the hydrogen emission spectrum and the double-slit experiment, then offers a meta-model grounded in finite measurement and nonlinear dynamics. The meta-model, Geofinitism, treats every symbol as a lossy compression of an interactional identity, the complete measured process from source through apparatus to registration. Its purpose is not to replace existing physical models but to keep the act of measurement visible, so that the language of science remains aligned with the finite distinctions from which it begins.

The work is offered as a tool for those who wish to examine the trajectories taken by scientific language and the trajectories left unexplored.

Chapter 1

The Contemporary Framing of Light

The present understanding of light rests on a sequence of experimental results and theoretical constructions that together form a coherent but still incomplete picture.

Photons are massless, relativistic quanta of the electromagnetic field. Because they travel at the speed of light they possess no rest mass and cannot be assigned a position without being destroyed. The ordinary Schrödinger equation, formulated for massive particles, therefore does not apply. A different theoretical structure is required: quantum electrodynamics, in which the electromagnetic field itself is quantised and photons appear as its excitations.

When a gas of hydrogen atoms is excited, electrons move into higher energy states and subsequently return to lower states, releasing electromagnetic radiation. The released radiation appears as a discrete set of wavelengths, the emission spectrum of atomic hydrogen. Each downward transition is associated with the emission of one photon whose energy equals the difference between the two levels. In the laboratory the spectrum is recorded as a pattern of bright lines against a darker background. The colour perceived by the eye is a combination of several such lines rather than any single wavelength.

The same quantised character is demonstrated in the double-slit experiment performed with extremely weak light. Individual detection events register as discrete points on a screen or camera array. When many such events are superimposed, an interference pattern emerges. The individual registrations appear particle-like; the accumulated pattern appears wave-like. If detectors are placed at the slits themselves, the interference pattern disappears and the detections split roughly equally between the two paths. The outcome depends on which measurement is performed.

Quantum electrodynamics accounts for these results by treating photons as quantised modes of the electromagnetic field. The field is regarded as more fundamental; the classical continuous wave of Maxwell's theory is understood as a statistical average over large numbers of quanta. The correspondence principle states that in the limit of large quantum numbers the predictions of the quantum theory coincide with those of classical physics. Decoherence, the irreversible interaction of a quantum system with its environment, is invoked to explain the rapid appearance of classical behaviour.

Difficulties remain. No non-relativistic wave function can be defined for a single photon that possesses all the familiar properties of massive-particle wave functions. The spatial extent associated with a visible-light wavelength is many orders of magnitude larger than the size of an electron, yet the interaction between them is routinely described. Experimental constraints show that a photon cannot be localised more tightly than a volume of order the cube of its wavelength. Some analyses therefore prefer the language of localised wave-corpuscles rather than point particles. An unresolved debate continues

over whether fields or particles should be regarded as the more fundamental description.

In this framing, light is neither purely wave nor purely particle. It is a relativistic epiphenomenon of a quantised field, registered as discrete events whose statistical distribution is described by wave equations. The language used to speak of these events, however, continues to rely on nouns that suggest static entities: the photon, the wavelength, the emission line, the interference fringe.

Chapter 2

Historical Trajectory of the Framing

The contemporary language did not appear fully formed. It is the product of successive decisions about which distinctions to treat as primary.

In the seventeenth century the Royal Society programme favoured the division of phenomena into discrete, measurable objects. Newton's corpuscular theory of light fitted this programme: light consisted of streams of particles travelling in straight lines. The wave theory that followed, developed by Young, Fresnel and Maxwell, replaced corpuscles with continuous fields oscillating in a medium. The medium itself was later abandoned, yet the continuous wave remained.

Planck's introduction of the quantum of action in 1900, and Einstein's subsequent use of that quantum to explain the photoelectric effect, restored a discrete element. The photon concept allowed the energy of light to be treated as packaged in finite units. Quantum electrodynamics, completed in the mid-twentieth century, absorbed both the continuous field and the discrete quanta into a single mathematical structure. Photons became excitations of the field; the classical wave became a limiting case of many quanta.

At each stage the language of science made a cut. Continuous intensity distributions were divided into lines; lines were further divided into photons; photons were later re-described as field modes. Each cut removed history and dynamics, converting a measured process into a named entity that could be manipulated as if it existed independently of the measurement that produced it. The goal remained the isolation of stable nouns, even while the underlying theory moved steadily toward fields, probabilities and relational descriptions.

Chapter 3

The Meta-Model of Measurement

Geofinitism begins from a different premise. All knowledge is built on measurements. Every measurement yields a finite distinction registered at the alphonic limit of an apparatus, the smallest separation the apparatus can reliably resolve. That distinction is an act of cutting: a continuous dynamical process is divided into two regions, and one region is selected for attention while the other is set aside.

The symbol assigned to the selected region is a lossy compression. History, provenance and the continuous dynamics that preceded the cut are discarded. The resulting noun, photon, wavelength, emission line, is treated as a static entity even though it is the label for a process that has already been truncated.

The primary unit of analysis is therefore not the noun but the interactional identity: the complete measured process that includes source, apparatus, registration and resolution. The interactional identity is the pattern of distinctions actually recorded. Derived models, photons, waves, transitions, are secondary compressions of that pattern. They remain useful, yet they are not more fundamental than the pattern itself.

A practical notation keeps the measurement process visible. Instead of the equality sign that asserts identity, a vertical bar asserts mapping under stated conditions:

$$\text{Hs}(\Delta t, \alpha) \mid I(x, y, z, t) + \varepsilon(\alpha) \quad (3.1)$$

Here Hs stands for the hydrogen interactional identity, Δt the integration time of the detector, α the alphonic limit, I the recorded grid of intensities, and ε the uncertainty associated with that limit. The bar does not claim that the spectrum “is” the grid; it records that the spectrum is measured as the grid under the stated conditions.

Chapter 4

Application to the Framing of Light

When the contemporary framing is read through this meta-model, the experimental results remain unchanged while the linguistic status of the nouns is altered.

The hydrogen emission spectrum is first a grid of intensities recorded over a finite integration time at a finite spatial and spectral resolution. Regions of higher activation are identified by a cut and labelled emission lines. The ratios of their positions are measured quantities. Wavelengths are then derived by applying a wave model to those ratios. The derivation is useful and accurate, yet it is a compression of the original grid.

The double-slit apparatus is a single detector system. It produces a statistical pattern of discrete registrations. When the pattern is examined as a whole it exhibits interference structure. When individual registrations are examined they appear localised. The two descriptions are alternative compressions of the same measured pattern. The question whether light “is” a wave or a particle arises only after the pattern has already been cut into two incompatible noun categories.

The photon itself is the symbol assigned after the fact to each registered distinction. It is neither a pre-existing particle nor a pre-existing wave. It is the label that allows the measured event to enter language and calculation. Quantum electrodynamics remains an extraordinarily successful compression of the patterns of distinction; the meta-model simply records that it is a compression.

Chapter 5

Language as Dynamical System

The same meta-model applies to language itself. Sequences of symbols are discrete samples of continuous cognitive and social dynamics. Takens' delay-coordinate embedding theorem shows that, under suitable conditions, the topology of an underlying continuous attractor can be reconstructed from such discrete samples. Language may therefore be treated as a time series generated by a high-dimensional dynamical system whose attractors correspond to stable meanings and whose trajectories correspond to the unfolding of discourse.

The practical consequence is that the refinement of scientific language is not merely a matter of inventing better nouns. It is a matter of choosing cuts that preserve more of the dynamical process, of marking compressions explicitly, and of extending the alphonic limit of the measurement systems, including the measurement systems of language itself.

Chapter 6

Consequences for Scientific Practice

Adoption of the meta-model does not require the abandonment of existing equations or experimental techniques. It requires a change in the order of presentation and in the awareness with which language is used.

Papers and textbooks begin with the measured pattern and the stated alphonic limit. Derived quantities are introduced as compressions. A Semantic Uncertainty Appendix records the cuts that have been made, the resolution of the apparatus, and the domain of validity of each model. Success is judged by the capacity of the language to support finer measurements, wider consensus and more precise interaction with the exogenous world.

Chapter 7

Closing Trajectory

The language of light has travelled a long path from corpuscles to fields. At each stage new distinctions were registered and new compressions invented. The meta-model offered here does not close that path. It supplies a second lens that keeps the act of measurement visible, so that future refinements of language remain grounded in the finite distinctions from which all scientific knowledge begins.

The work continues wherever new measurements are made and new symbols are assigned.

Appendix A

Sample Rewritings

Hydrogen Spectrum

$$\text{Hs}(\Delta t, \alpha) \mid I(x, y, z, t) + \varepsilon(\alpha) \quad (\text{A.1})$$

Regions of higher intensity are identified by a cut:

$$L_i(\text{Hs}, \Delta t, \alpha) \mid \text{subset of } I(x, y, z, t) \quad (\text{A.2})$$

Ratios of positions:

$$R_{ij}(\text{Hs}, \Delta t, \alpha) \mid \text{ratio of positions of } L_i \text{ and } L_j \quad (\text{A.3})$$

Derived wavelengths (compression):

$$\lambda_i(\text{Hs}, \Delta t, \alpha) \mid \text{derived from } R_{ij} \text{ using the wave model} \quad (\text{A.4})$$

Double-Slit Pattern

$$\text{DS}(\Delta t, \alpha) \mid I(x, y, t) + \varepsilon(\alpha) \quad (\text{A.5})$$

where DS denotes the double-slit interactional identity.

Appendix B

Template for a Semantic Uncertainty Appendix

- Measured pattern: description of the raw grid or distribution.
- Alphonic limit α : resolution of the apparatus (spatial, spectral, temporal).
- Cuts made: explicit list of distinctions imposed (e.g., line versus background, particle-like versus wave-like).
- Derived quantities: list of compressions and the models used to obtain them.
- Domain of validity: conditions under which each compression remains useful.
- Uncertainty $\varepsilon(\alpha)$: quantitative or qualitative statement of residual uncertainty.